KERALA POLYTECHNIC · REVISION 2026

# Computer Networks II

Complete student lesson for course code **4151**.

Revision 2026 Semester 4 Program Core 4 modules

## Course snapshot

**Scheme:** 4 (L:3,T:1,P:0); 60 periods

**Credits:** 4

**Contact:** 60 periods per semester

**Module hours listed:** 58

## Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

**Source handling:** Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

**Transparency note:** Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

## How to Study This Lesson

### Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

**Study principle:** Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

**Handbook method:** Do not treat a topic as completed after reading its heading. For every topic card, complete the learning objectives, follow the conceptual framework, write the worked answer method, and answer the self-check questions. This creates a study record that is useful for class learning and examination preparation.

## Course Dashboard

Course code

**4151**

Semester

**4**

Credits

**4**

Teaching scheme

**4 (L:3,T:1,P:0); 60 periods**

**Module-hour and outcome mapping**

| No. | Module | Official hours | Relevant CO / level |
|---|---|---|---|
| 1 | Network control devices, wireless and services | 14 | CO1 |
| 2 | Active Directory and security | 14 | CO2 |

| No. | Module | Official hours | Relevant CO / level |
|---|---|---|---|
| 3 | IOS, switches and router configuration | 16 | CO3 |
| 4 | Static and dynamic routing | 14 | CO4 |

### Prerequisites

- Computer Networks I and basic TCP/IP addressing.

### How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

## Objectives, Course Outcomes and CO–PO Information

### Official course objectives

1. Outline network control devices and IEEE standards.
2. Demonstrate active directory and security.
3. Illustrate fundamentals of IOS, switches and router configuration.
4. Outline static and dynamic routing protocols.

### Course Outcomes

1. Compare IPv4 and IPv6 and explain network-control devices and standards.
2. Demonstrate domain, Active Directory and security concepts.
3. Illustrate IOS, switch and router fundamentals and configuration.
4. Outline static and dynamic routing protocols.

**CO–PO mapping as supplied**

CO–PO mapping is reproduced from the supplied syllabus. The numerical mapping is retained as printed; blank cells are not silently converted into values.

Legend: 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped.

# Complete Syllabus Coverage

## Official syllabus point-by-point coverage

This audit layer maps every official source block to the corresponding lesson module. The source transcript is retained verbatim as extracted from the supplied syllabus; the study guidance below is educational support and is not presented as additional official syllabus content. No official point is intentionally omitted.

**Source-to-lesson coverage map**

| Module | Lesson topic cards | Source basis | Official point units |
|---|---|---|---|
| Module 1 | 5 | Official Contents block | 5 |
| Module 2 | 4 | Official Contents block | 2 |
| Module 3 | 6 | Official Contents block | 6 |
| Module 4 | 6 | Official Contents block | 6 |

Use this checklist to confirm that every official module topic is represented in the lesson.

**Complete official topic coverage checklist**

| Module | Topic code | Topic | Hours |
|---|---|---|---|
| module-1 | M1.01 | IPv4 and IPv6 addressing | 1 |
| module-1 | M1.02 | Network-control devices | 3 |
| module-1 | M1.03 | Wireless technologies | 3 |
| module-1 | M1.04 | IEEE 802 standards | 3 |
| module-1 | M1.05 | DNS and DHCP | 4 |
| module-2 | M2.01 | Domain controllers and Active Directory | 3 |
| module-2 | M2.02 | Domain and workgroup users | 4 |
| module-2 | M2.03 | File and folder security | 4 |

| Module | Topic code | Topic | Hours |
|---|---|---|---|
| module-2 | M2.04 | Security, firewall and backup | 3 |
| module-3 | M3.01 | Switch and router hardware and memory | 2 |
| module-3 | M3.02 | Router and switch interfaces | 2 |
| module-3 | M3.03 | Internetwork Operating System | 3 |
| module-3 | M3.04 | Basic switch configuration | 3 |
| module-3 | M3.05 | Basic router configuration | 4 |
| module-3 | M3.06 | Registers, clock and logging | 2 |
| module-4 | M4.01 | IOS upgrading and disaster recovery | 2 |
| module-4 | M4.02 | Authentication and accounting | 1 |
| module-4 | M4.03 | TCP/IP static and dynamic routing | 3 |
| module-4 | M4.04 | IGP and EGP | 4 |
| module-4 | M4.05 | Protocol choice and routing information | 2 |
| module-4 | M4.06 | Route control and redistribution | 2 |

## Learning Roadmap

**1. Network control devices, wireless and services**
**CO1 · 14 hours**

**2. Active Directory and security**
**CO2 · 14 hours**

**3. IOS, switches and router configuration**
**CO3 · 16 hours**

**4. Static and dynamic routing**
**CO4 · 14 hours**

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

# Network control devices, wireless and services

Network infrastructure uses addressing, switching, routing, wireless technologies and support services. DNS maps names to addresses; DHCP supplies configuration automatically.

**Hours: 14**

**CO1 · Bloom level: Understanding**

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**Handbook study routine:** Complete Module 1 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

## Official source coverage for Module 1

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. Familiarization of networking Control Devices. Current networking scenario- Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).
IEEE standards 802.X, 802.XX
Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of DNS and DHCP

## Point-by-point study guide

**Official syllabus point 1: Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask.**

#### Exact source point

**Syllabus text:** Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask.

#### How to understand and study it

This point is an official syllabus requirement: “Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ Compare IP addresses-IPV4, IPV6:--Understanding subnet mask. □□□□ technical terms English-□□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. using the official terminology retained above.
- Organise the important elements of Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. into a logical sequence, classification, structure, or calculation.
- Connect Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask.?
- How would you distinguish Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. from a closely related idea?
- Where would an engineer or technician encounter Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask., and what must be checked before using it?
- What common mistake would make an answer or operation involving Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Compare IP addresses-IPV4 and IPV6:--Understanding subnet mask. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

### Official syllabus point 2: Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).

#### Exact source point

**Syllabus text:** Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).

#### How to understand and study it

This point is an official syllabus requirement: “Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).”. Study the sequence from input or initial condition to operation and final output. Note the role of each named stage and the cause-and-effect relationship. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Router- Repeaters, VPN devices, Modem- wireless network technologies (Wi-Fi ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). must be learned as both a concept and a calculation procedure. The student should know what each quantity represents, the conditions under which a relation is valid, the unit of every quantity, and how to check whether the final answer is physically reasonable.

## Learning objectives

- Define and scope Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). using the official terminology retained above.
- Organise the important elements of Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). into a logical sequence, classification, structure, or calculation.
- Connect Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).. It is a study organisation method, not an additional official syllabus requirement.

- Identify the given quantities and write them with symbols and SI units.
- Select the governing definition, equation, graph, or relationship instead of starting with arithmetic.

- Substitute values only after checking units and the stated operating condition.
- Interpret the result in words and check its sign, magnitude, unit, and limiting behaviour.

### Engineering application lens

In an engineering setting, Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). is useful when a designer or technician must convert an observation into a measurable value, predict a response, compare alternatives, or verify that a component is operating within its rated condition. The practical question is always: what is measured or known, what is required, what model is appropriate, and how will the result be checked?

### Worked answer method

**Given:** the quantities and conditions stated in the question.
**Required:** the unknown quantity, including its unit.
**Calculation:** write the governing relation symbolically, substitute consistently converted values, and show the unit cancellation.
**Answer:** report the result with a suitable number of significant figures and one sentence of interpretation.

### Exam answer blueprint

For a short-answer question on Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max)., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max)., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Familiarization of networking Control Devices. Current

networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max).?

- How would you distinguish Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). from a closely related idea?
- Where would an engineer or technician encounter Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max)., and what must be checked before using it?
- What common mistake would make an answer or operation involving Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). incorrect?

### Common mistakes and safety emphasis

- Using a memorised relation without checking its conditions.
- Mixing prefixes or units, such as milli, kilo, mega, seconds, minutes, or degrees.
- Reporting a numerical value without a unit or without explaining what it means.
- Rounding intermediate values so aggressively that the final answer changes.

**Malayalam support:** Familiarization of networking Control Devices. Current networking scenario-Commonly used networking devices- Switch, Hub and Router- Repeaters, VPN devices and Modem- wireless network technologies (Wi-Fi, Bluetooth, Wi-max). □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

## Official syllabus point 3: IEEE standards 802.X, 802.XX

### Exact source point

**Syllabus text:** IEEE standards 802.X, 802.XX

### How to understand and study it

This point is an official syllabus requirement: “IEEE standards 802.X, 802.XX”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ IEEE standards 802.X, 802.XX ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

IEEE standards 802.X, 802.XX is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope IEEE standards 802.X, 802.XX using the official terminology retained above.
- Organise the important elements of IEEE standards 802.X, 802.XX into a logical sequence, classification, structure, or calculation.
- Connect IEEE standards 802.X, 802.XX with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on IEEE standards 802.X, 802.XX with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading IEEE standards 802.X, 802.XX. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of IEEE standards 802.X, 802.XX is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on IEEE standards 802.X, 802.XX, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is IEEE standards 802.X, 802.XX, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining IEEE standards 802.X, 802.XX?
- How would you distinguish IEEE standards 802.X, 802.XX from a closely related idea?
- Where would an engineer or technician encounter IEEE standards 802.X, 802.XX, and what must be checked before using it?
- What common mistake would make an answer or operation involving IEEE standards 802.X, 802.XX incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** IEEE standards 802.X, 802.XX [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

### Official syllabus point 4: Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of

#### Exact source point

> **Syllabus text:** Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of

#### How to understand and study it

This point is an official syllabus requirement: “Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Write the operating sequence in numbered steps, including the input, intermediate action, and output.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ Concept of DNS DHCP server -Working of DNS, DHCP-Installation requirement of □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of using the official terminology retained above.
- Organise the important elements of Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of into a logical sequence, classification, structure, or calculation.
- Connect Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of?
- How would you distinguish Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of from a closely related idea?
- Where would an engineer or technician encounter Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of, and what must be checked before using it?
- What common mistake would make an answer or operation involving Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Concept of DNS DHCP server -Working of DNS and DHCP-Installation requirement of □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

## Official syllabus point 5: DNS and DHCP

### Exact source point

**Syllabus text:** DNS and DHCP

### How to understand and study it

This point is an official syllabus requirement: “DNS and DHCP”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ technical terminology English-□□□□□□□□ □□□□□□□□□□□. □□□□□ concept □□□□□□□□□□□□, □□□□ definition, working or procedure, application, exam answer □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

DNS and DHCP is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope DNS and DHCP using the official terminology retained above.
- Organise the important elements of DNS and DHCP into a logical sequence, classification, structure, or calculation.
- Connect DNS and DHCP with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on DNS and DHCP with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading DNS and DHCP. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of DNS and DHCP is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on DNS and DHCP, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is DNS and DHCP, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining DNS and DHCP?
- How would you distinguish DNS and DHCP from a closely related idea?
- Where would an engineer or technician encounter DNS and DHCP, and what must be checked before using it?
- What common mistake would make an answer or operation involving DNS and DHCP incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

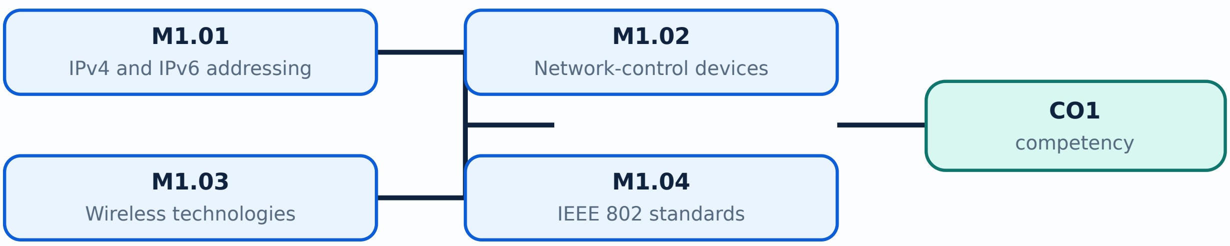
**Malayalam support:** DNS and DHCP ▯▯▯▯ topic ▯▯▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ exact technical term ▯▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯▯ principle, named parts/steps, application, limitation ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. English terminology, symbols, units, diagram labels ▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯. Exam answer ▯▯▯▯▯▯▯▯▯▯▯ concept-▯▯▯▯▯ ▯▯▯▯▯▯–▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯▯.

## Learning goals

- Compare IPv4 and IPv6.
- Identify network-control devices.
- Explain wireless standards, DNS and DHCP.

## Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.




Learning map for Module module-1: the listed topics build the module competency.

## M1.01 — IPv4 and IPv6 addressing (1 hours)

### Concept and explanation

IPv4 uses 32-bit addresses and subnet masks; IPv6 uses 128-bit addresses and a larger address space. Prefix notation identifies the network portion in IPv6 and modern IPv4 practice.

**Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

### Study points

- Compare address length.
- Explain subnet-mask purpose.
- Recognise basic notation.

**Engineering / workplace application:** Address planning is essential for routing and network management.

**Common mistake:** An IPv6 address is not simply an IPv4 address with extra decimal digits; it uses hexadecimal groups and prefix notation.

## Handbook treatment

M1.01 — IPv4 and IPv6 addressing (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M1.01 — IPv4 and IPv6 addressing (1 hours) using the official terminology retained above.
- Organise the important elements of M1.01 — IPv4 and IPv6 addressing (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.01 — IPv4 and IPv6 addressing (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on M1.01 — IPv4 and IPv6 addressing (1 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M1.01 — IPv4 and IPv6 addressing (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M1.01 — IPv4 and IPv6 addressing (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M1.01 — IPv4 and IPv6 addressing (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M1.01 — IPv4 and IPv6 addressing (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.01 — IPv4 and IPv6 addressing (1 hours)?
- How would you distinguish M1.01 — IPv4 and IPv6 addressing (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.01 — IPv4 and IPv6 addressing (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.01 — IPv4 and IPv6 addressing (1 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M1.01 — IPv4 and IPv6 addressing (1 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M1.02 — Network-control devices (3 hours)

### Concept and explanation

Hubs repeat signals, switches forward frames using MAC addresses, routers forward packets between networks, repeaters regenerate signals, VPN devices create protected tunnels and modems adapt signals to a service.

**Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□ □□□□□□□□.

### Study points

- State device functions.
- Compare switch and router.
- Select a device for a scenario.

**Engineering / workplace application:** Correct device selection supports scalable campus and enterprise networks.

# Handbook treatment

M1.02 — Network-control devices (3 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope M1.02 — Network-control devices (3 hours) using the official terminology retained above.
- Organise the important elements of M1.02 — Network-control devices (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.02 — Network-control devices (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on M1.02 — Network-control devices (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M1.02 — Network-control devices (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, M1.02 — Network-control devices (3 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on M1.02 — Network-control devices (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M1.02 — Network-control devices (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.02 — Network-control devices (3 hours)?
- How would you distinguish M1.02 — Network-control devices (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.02 — Network-control devices (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.02 — Network-control devices (3 hours) incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** M1.02 — Network-control devices (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M1.03 — Wireless technologies (3 hours)

### Concept and explanation

Wi-Fi, Bluetooth and WiMAX provide wireless connectivity with different range, data rate, power and topology characteristics. Security and interference must be considered.

> **Malayalam support:** [illegible] English technical terms, symbols, units, equations, and standard abbreviations [illegible].

### Study points

- Compare wireless technologies.
- State application examples.
- Mention basic interference and security concerns.

## Handbook treatment

M1.03 — Wireless technologies (3 hours) should be followed as a signal or information path. Explain what is generated, how it is modified or transmitted, what can disturb it, and how the receiver reconstructs or uses it.

### Learning objectives

- Define and scope M1.03 — Wireless technologies (3 hours) using the official terminology retained above.
- Organise the important elements of M1.03 — Wireless technologies (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.03 — Wireless technologies (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on M1.03 — Wireless technologies (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M1.03 — Wireless technologies (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the information-bearing signal and the relevant source or channel.
- Describe the blocks or stages in order.
- Explain the role of bandwidth, noise, attenuation, distortion, coding, modulation, or protocol rules where applicable.
- Compare performance using range, capacity, reliability, complexity, cost, and application.

### Engineering application lens

Engineering systems use M1.03 — Wireless technologies (3 hours) when information must be transferred reliably between equipment, instruments, controllers, or users. The choice of method is a balance between signal quality, distance, data rate, interference, infrastructure, safety, and maintenance.

### Worked answer method

**Given:** source, channel, receiver, signal condition, or communication requirement.
**Required:** block diagram, operating explanation, comparison, or fault analysis.
**Method:** trace the signal from source to destination and mark where conversion, loss, interference, or recovery occurs.
**Answer:** state the principle, blocks, performance factors, application, and limitation.

### Exam answer blueprint

For a short-answer question on M1.03 — Wireless technologies (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M1.03 — Wireless technologies (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.03 — Wireless technologies (3 hours)?
- How would you distinguish M1.03 — Wireless technologies (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.03 — Wireless technologies (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.03 — Wireless technologies (3 hours) incorrect?

### Common mistakes and safety emphasis

- Using transmitter, channel, and receiver as labels without explaining their functions.
- Confusing bandwidth, data rate, frequency, and range.
- Ignoring noise, attenuation, interference, synchronization, or protocol compatibility.
- Comparing systems without using common criteria.

**Malayalam support:** M1.03 — Wireless technologies (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M1.04 — IEEE 802 standards (3 hours)

### Concept and explanation

IEEE 802 standards define families of LAN and wireless networking technologies. The exact standard determines medium access, physical layer and interoperability requirements.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□.

### Study points

- Explain the role of standards.
- Relate 802 standards to LAN/WLAN.
- Use standard terminology accurately.

# Handbook treatment

M1.04 — IEEE 802 standards (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M1.04 — IEEE 802 standards (3 hours) using the official terminology retained above.
- Organise the important elements of M1.04 — IEEE 802 standards (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.04 — IEEE 802 standards (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on M1.04 — IEEE 802 standards (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M1.04 — IEEE 802 standards (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M1.04 — IEEE 802 standards (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M1.04 — IEEE 802 standards (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M1.04 — IEEE 802 standards (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.04 — IEEE 802 standards (3 hours)?
- How would you distinguish M1.04 — IEEE 802 standards (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.04 — IEEE 802 standards (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.04 — IEEE 802 standards (3 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M1.04 — IEEE 802 standards (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M1.05 — DNS and DHCP (4 hours)

### Concept and explanation

DNS resolves names through a distributed hierarchy. DHCP leases configuration such as IP address, mask, gateway and DNS server. Installation requires correct server roles, scopes, records and client settings.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□□ □□□□□□□□□□.

### Study points

- Trace a DNS lookup.
- Outline a DHCP lease.
- List installation requirements.

> **Engineering / workplace application:** DNS and DHCP reduce manual configuration in enterprise networks.

> **Common mistake:** DNS resolution failure and DHCP address exhaustion are different faults and require different diagnostics.

# Handbook treatment

M1.05 — DNS and DHCP (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope M1.05 — DNS and DHCP (4 hours) using the official terminology retained above.
- Organise the important elements of M1.05 — DNS and DHCP (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M1.05 — DNS and DHCP (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Network control devices, wireless and services.
- Write an examination answer on M1.05 — DNS and DHCP (4 hours) with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading M1.05 — DNS and DHCP (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of M1.05 — DNS and DHCP (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M1.05 — DNS and DHCP (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M1.05 — DNS and DHCP (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M1.05 — DNS and DHCP (4 hours)?
- How would you distinguish M1.05 — DNS and DHCP (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M1.05 — DNS and DHCP (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M1.05 — DNS and DHCP (4 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M1.05 — DNS and DHCP (4 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□□.

**Module summary:** Network operation depends on correctly combining addresses, devices, standards and automatic configuration services.

OFFICIAL MODULE 2

# Active Directory and security

Domain-based administration centralises identity, policy and resource access. File permissions, backups, firewalls and encryption protect availability and confidentiality.

**Hours: 14**
**CO2 · Bloom level: Understanding**

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**Handbook study routine:** Complete Module 2 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

### Official source coverage for Module 2

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

: Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.

## Point-by-point study guide

## Official syllabus point 1: Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network.

### Exact source point

**Syllabus text:** Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network.

### How to understand and study it

This point is an official syllabus requirement: “Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network.”. Study this as a structure: identify each named part, state its function, and explain how the parts are connected or arranged. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ Domain controllers, Active directory. Domain, workgroup network architecture - client server architecture -the requirements of domain network. □□□□ technical terms English-□□□□□□□□ □□□□□□□□. □□□□□ definition □□□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. using the official terminology retained above.
- Organise the important elements of Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. into a logical sequence, classification, structure, or calculation.
- Connect Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network.. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network.?
- How would you distinguish Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. from a closely related idea?
- Where would an engineer or technician encounter Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network., and what must be checked before using it?
- What common mistake would make an answer or operation involving Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** Domain controllers and Active directory. Domain and workgroup network architecture - client server architecture -the requirements of domain network. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

**Official syllabus point 2: Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.**

### Exact source point

> **Syllabus text:** Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.

### How to understand and study it

This point is an official syllabus requirement: “Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. Draw a labelled arrangement or block diagram and write the function of every labelled part.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ Domain, workgroup users:-Managing user, group accounts-Adding group memberships-Physical, logical components of domain-Understanding child domain-additional domain controller. Securing files □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. using the official terminology retained above.
- Organise the important elements of Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. into a logical sequence, classification, structure, or calculation.
- Connect Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup.?
- How would you distinguish Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. from a closely related idea?
- Where would an engineer or technician encounter Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup., and what must be checked before using it?
- What common mistake would make an answer or operation involving Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting

share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

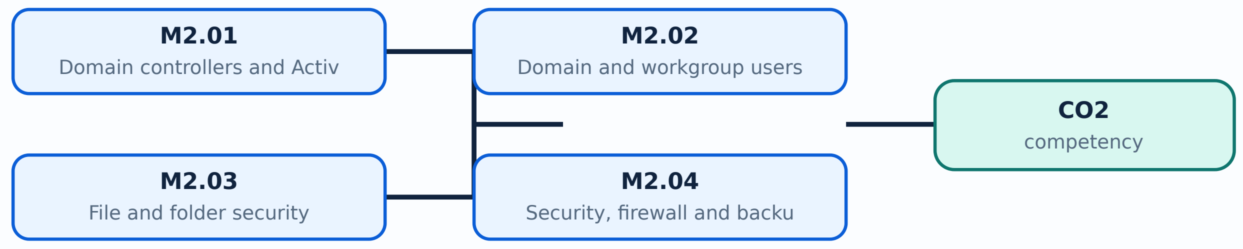
**Malayalam support:** Domain and workgroup users:-Managing user, group accounts-Adding group memberships-Physical and logical components of domain-Understanding child domain-additional domain controller. Securing files and folders: - setting share permission-setting security permission. Security and encryption:-firewall-general concepts and types -system backup- active directory backup. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

## Learning goals

- Explain domain controllers and Active Directory.
- Manage domain/workgroup users conceptually.
- Apply permission and security principles.

## Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.




Learning map for Module module-2: the listed topics build the module competency.

## M2.01 — Domain controllers and Active Directory (3 hours)

### Concept and explanation

A domain controller authenticates users and applies directory-based administration. Active Directory stores objects, accounts, groups and policy information in a structured domain.

**Malayalam support:** [illegible] English technical terms, symbols, units, equations, and standard abbreviations [illegible].

### Study points

- Define domain controller.
- State Active Directory functions.
- Compare centralised and local administration.

## Handbook treatment

M2.01 — Domain controllers and Active Directory (3 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

### Learning objectives

- Define and scope M2.01 — Domain controllers and Active Directory (3 hours) using the official terminology retained above.
- Organise the important elements of M2.01 — Domain controllers and Active Directory (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.01 — Domain controllers and Active Directory (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on M2.01 — Domain controllers and Active Directory (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M2.01 — Domain controllers and Active Directory (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

### Engineering application lens

Engineering use of M2.01 — Domain controllers and Active Directory (3 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

### Worked answer method

**Given:** the quantity to be sensed, the operating environment, and the required decision or control action.
**Required:** a suitable measurement chain or an explanation of how the instrument operates.
**Method:** map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
**Answer:** state the measured quantity, principle, important characteristics, application, and limitation.

### Exam answer blueprint

For a short-answer question on M2.01 — Domain controllers and Active Directory (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M2.01 — Domain controllers and Active Directory (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.01 — Domain controllers and Active Directory (3 hours)?
- How would you distinguish M2.01 — Domain controllers and Active Directory (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.01 — Domain controllers and Active Directory (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.01 — Domain controllers and Active Directory (3 hours) incorrect?

### Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

**Malayalam support:** M2.01 — Domain controllers and Active Directory (3 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□.

### M2.02 — Domain and workgroup users (4 hours)

#### Concept and explanation

A workgroup manages accounts locally, while a domain centralises authentication and policy. User and group accounts, memberships, child domains and additional domain controllers support administration and redundancy.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□ □□□□□□□□.

#### Study points

- Compare domain and workgroup.
- Explain group membership.
- Identify physical and logical components.

## Handbook treatment

M2.02 — Domain and workgroup users (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M2.02 — Domain and workgroup users (4 hours) using the official terminology retained above.
- Organise the important elements of M2.02 — Domain and workgroup users (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.02 — Domain and workgroup users (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on M2.02 — Domain and workgroup users (4 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M2.02 — Domain and workgroup users (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M2.02 — Domain and workgroup users (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M2.02 — Domain and workgroup users (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M2.02 — Domain and workgroup users (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.02 — Domain and workgroup users (4 hours)?
- How would you distinguish M2.02 — Domain and workgroup users (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.02 — Domain and workgroup users (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.02 — Domain and workgroup users (4 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M2.02 — Domain and workgroup users (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

### M2.03 — File and folder security (4 hours)

#### Concept and explanation

Share permissions and security permissions control access at different layers. Effective access depends on the combined rules, inheritance, ownership and principle of least privilege.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□ □□□□□□□□.

#### Study points

- Differentiate share and security permission.
- Explain inheritance.
- Apply least-privilege reasoning.

> **Engineering / workplace application:** File servers require documented permissions and audit trails.

# Handbook treatment

M2.03 — File and folder security (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M2.03 — File and folder security (4 hours) using the official terminology retained above.
- Organise the important elements of M2.03 — File and folder security (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.03 — File and folder security (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on M2.03 — File and folder security (4 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M2.03 — File and folder security (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M2.03 — File and folder security (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M2.03 — File and folder security (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M2.03 — File and folder security (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.03 — File and folder security (4 hours)?
- How would you distinguish M2.03 — File and folder security (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.03 — File and folder security (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.03 — File and folder security (4 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M2.03 — File and folder security (4 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M2.04 — Security, firewall and backup (3 hours)

### Concept and explanation

A firewall filters traffic according to rules and network context. Encryption protects data confidentiality; system and Active Directory backups support recovery from failure or attack.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

### Study points

- Classify firewall concepts.
- Explain backup purpose.
- Relate security controls to threats.

> **Engineering / workplace application:** Security is a layered process rather than a single firewall setting.

## Handbook treatment

M2.04 — Security, firewall and backup (3 hours) must be understood as a control-of-risk topic. Identify the hazard, the possible harm, the conditions that increase the risk, and the hierarchy of controls that prevents the incident.

### Learning objectives

- Define and scope M2.04 — Security, firewall and backup (3 hours) using the official terminology retained above.
- Organise the important elements of M2.04 — Security, firewall and backup (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M2.04 — Security, firewall and backup (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Active Directory and security.
- Write an examination answer on M2.04 — Security, firewall and backup (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M2.04 — Security, firewall and backup (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Identify the source of energy or unsafe condition.
- Describe the possible consequence and the people, equipment, or environment exposed.
- Apply elimination, substitution, engineering control, administrative control, and personal protective equipment in that order.
- State inspection, isolation, emergency response, reporting, and recovery requirements where relevant.

### Engineering application lens

In engineering practice, M2.04 — Security, firewall and backup (3 hours) is part of normal design, operation, maintenance, and troubleshooting—not a separate final paragraph. The safest answer connects the control measure to the hazard and explains why the measure is effective.

### Worked answer method

**Given:** the work activity, equipment, hazard, or incident condition.
**Required:** identify risk and propose controls.
**Method:** describe hazard → consequence → control → verification → emergency action.
**Answer:** give specific controls, responsible action, and one condition that must be checked before work begins.

### Exam answer blueprint

For a short-answer question on M2.04 — Security, firewall and backup (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M2.04 — Security, firewall and backup (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M2.04 — Security, firewall and backup (3 hours)?
- How would you distinguish M2.04 — Security, firewall and backup (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M2.04 — Security, firewall and backup (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M2.04 — Security, firewall and backup (3 hours) incorrect?

### Common mistakes and safety emphasis

- Naming PPE as the only control when isolation or guarding is required.
- Describing a hazard without stating the consequence.
- Ignoring stored energy, restart, access, housekeeping, or emergency isolation.
- Treating a safety instruction as optional or disconnected from the operating procedure.

**Malayalam support:** M2.04 — Security, firewall and backup (3 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

**Module summary:** Identity, permissions and recovery controls protect network resources and users.

OFFICIAL MODULE 3

# IOS, switches and router configuration

Cisco IOS-based devices use hardware, interfaces, memory and command-line configuration. Safe administration requires saving, backing up, restoring and logging configuration changes.

**Hours: 16**
**CO3 · Bloom level: Understanding**

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**Handbook study routine:** Complete Module 3 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

## Official source coverage for Module 3

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces - Internetwork Operating System (IOS) - Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up Buffered Logging-Setting up Trap Logging.

## Point-by-point study guide

## Official syllabus point 1: Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces

### Exact source point

**Syllabus text:** Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces

### How to understand and study it

This point is an official syllabus requirement: “Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□ Introduction to Switch, Router. Switch, Router Hardware-Router, Switch Interfaces □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces using the official terminology retained above.
- Organise the important elements of Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces into a logical sequence, classification, structure, or calculation.
- Connect Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces?
- How would you distinguish Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces from a closely related idea?
- Where would an engineer or technician encounter Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces, and what must be checked before using it?
- What common mistake would make an answer or operation involving Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Introduction to Switch and Router. Switch and Router Hardware-Router and Switch Interfaces [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## Official syllabus point 2: Internetwork Operating System (IOS)

### Exact source point

**Syllabus text:** Internetwork Operating System (IOS)

### How to understand and study it

This point is an official syllabus requirement: “Internetwork Operating System (IOS)”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ Internetwork Operating System (IOS) □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Internetwork Operating System (IOS) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

## Learning objectives

- Define and scope Internetwork Operating System (IOS) using the official terminology retained above.
- Organise the important elements of Internetwork Operating System (IOS) into a logical sequence, classification, structure, or calculation.
- Connect Internetwork Operating System (IOS) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on Internetwork Operating System (IOS) with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading Internetwork Operating System (IOS). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

## Engineering application lens

For engineering practice, Internetwork Operating System (IOS) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

## Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on Internetwork Operating System (IOS), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Internetwork Operating System (IOS), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Internetwork Operating System (IOS)?
- How would you distinguish Internetwork Operating System (IOS) from a closely related idea?
- Where would an engineer or technician encounter Internetwork Operating System (IOS), and what must be checked before using it?
- What common mistake would make an answer or operation involving Internetwork Operating System (IOS) incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** Internetwork Operating System (IOS) □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□.

### Official syllabus point 3: Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and

#### Exact source point

**Syllabus text:** Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and

#### How to understand and study it

This point is an official syllabus requirement: “Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ Basic configurations of a Switch-setting, configuring commands– Saving, Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and using the official terminology retained above.
- Organise the important elements of Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and into a logical sequence, classification, structure, or calculation.
- Connect Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and?
- How would you distinguish Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and from a closely related idea?

- Where would an engineer or technician encounter Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and, and what must be checked before using it?
- What common mistake would make an answer or operation involving Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration-Configuring Router with <copy> and incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Basic configurations of a Switch-setting and configuring commands– Saving and Erasing Configurations-Memory on Routers-Basic Router Configuration- Configuring Router with <copy> and □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

## Official syllabus point 4: TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading

### Exact source point

**Syllabus text:** TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading

### How to understand and study it

This point is an official syllabus requirement: “TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading using the official terminology retained above.
- Organise the important elements of TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading into a logical sequence, classification, structure, or calculation.
- Connect TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading?
- How would you distinguish TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading from a closely related idea?
- Where would an engineer or technician encounter TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading, and what must be checked before using it?
- What common mistake would make an answer or operation involving TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** TFTP-Setting the Bootstrap Behavior-Configuration Register Settings-upgrading □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

### Official syllabus point 5: Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up

#### Exact source point

**Syllabus text:** Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up

#### How to understand and study it

This point is an official syllabus requirement: “Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□□ Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up using the official terminology retained above.
- Organise the important elements of Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up into a logical sequence, classification, structure, or calculation.
- Connect Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up?
- How would you distinguish Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up from a closely related idea?
- Where would an engineer or technician encounter Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up, and what must be checked before using it?
- What common mistake would make an answer or operation involving Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Router's IOS- Configuring the Router's Clock-IOS Message Logging.-Setting up □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

## Official syllabus point 6: Buffered Logging-Setting up Trap Logging.

### Exact source point

**Syllabus text:** Buffered Logging-Setting up Trap Logging.

### How to understand and study it

This point is an official syllabus requirement: “Buffered Logging-Setting up Trap Logging.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ Buffered Logging-Setting up Trap Logging. ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Buffered Logging-Setting up Trap Logging. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Buffered Logging-Setting up Trap Logging. using the official terminology retained above.
- Organise the important elements of Buffered Logging-Setting up Trap Logging. into a logical sequence, classification, structure, or calculation.
- Connect Buffered Logging-Setting up Trap Logging. with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on Buffered Logging-Setting up Trap Logging. with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Buffered Logging-Setting up Trap Logging.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Buffered Logging-Setting up Trap Logging. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Buffered Logging-Setting up Trap Logging., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Buffered Logging-Setting up Trap Logging., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Buffered Logging-Setting up Trap Logging.?
- How would you distinguish Buffered Logging-Setting up Trap Logging. from a closely related idea?
- Where would an engineer or technician encounter Buffered Logging-Setting up Trap Logging., and what must be checked before using it?
- What common mistake would make an answer or operation involving Buffered Logging-Setting up Trap Logging. incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

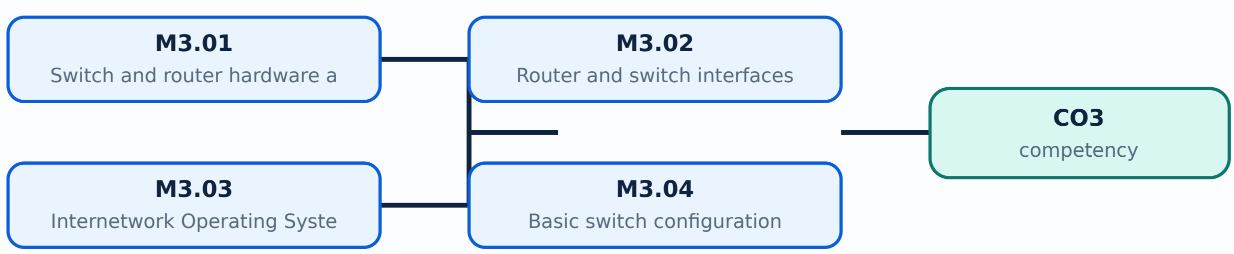
**Malayalam support:** Buffered Logging-Setting up Trap Logging. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

## Learning goals

- Identify router/switch hardware and memory.
- Explain interfaces and IOS.
- Outline basic switch and router configuration.

## Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.




Learning map for Module module-3: the listed topics build the module competency.

### M3.01 — Switch and router hardware and memory (2 hours)

#### Concept and explanation

Network devices contain processors, interfaces and memory areas such as running configuration, startup configuration and IOS storage. Understanding memory explains why changes may disappear after reboot.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

#### Study points

- Identify major hardware.
- Distinguish running and startup configuration.
- State backup relevance.

## Handbook treatment

M3.01 — Switch and router hardware and memory (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M3.01 — Switch and router hardware and memory (2 hours) using the official terminology retained above.
- Organise the important elements of M3.01 — Switch and router hardware and memory (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.01 — Switch and router hardware and memory (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.01 — Switch and router hardware and memory (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.01 — Switch and router hardware and memory (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M3.01 — Switch and router hardware and memory (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M3.01 — Switch and router hardware and memory (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.01 — Switch and router hardware and memory (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.01 — Switch and router hardware and memory (2 hours)?
- How would you distinguish M3.01 — Switch and router hardware and memory (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.01 — Switch and router hardware and memory (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.01 — Switch and router hardware and memory (2 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M3.01 — Switch and router hardware and memory (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M3.02 — Router and switch interfaces (2 hours)

### Concept and explanation

Interfaces connect a device to network media and require correct speed, duplex, addressing, status and descriptions. Switch ports and router interfaces have different forwarding roles.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□□ □□□□□□□□□□.

### Study points

- Describe interface parameters.
- Explain administrative and line protocol state.
- Use interface terminology.

## Handbook treatment

M3.02 — Router and switch interfaces (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M3.02 — Router and switch interfaces (2 hours) using the official terminology retained above.
- Organise the important elements of M3.02 — Router and switch interfaces (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.02 — Router and switch interfaces (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.02 — Router and switch interfaces (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.02 — Router and switch interfaces (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M3.02 — Router and switch interfaces (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M3.02 — Router and switch interfaces (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.02 — Router and switch interfaces (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.02 — Router and switch interfaces (2 hours)?
- How would you distinguish M3.02 — Router and switch interfaces (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.02 — Router and switch interfaces (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.02 — Router and switch interfaces (2 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M3.02 — Router and switch interfaces (2 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M3.03 — Internetwork Operating System (3 hours)

### Concept and explanation

IOS provides command modes, configuration commands, show commands and copy operations. Command hierarchy controls access to monitoring and configuration functions.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□□.

### Study points

- Identify basic IOS modes.
- Explain show and configuration commands.
- Use safe command sequencing.

## Handbook treatment

M3.03 — Internetwork Operating System (3 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope M3.03 — Internetwork Operating System (3 hours) using the official terminology retained above.
- Organise the important elements of M3.03 — Internetwork Operating System (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.03 — Internetwork Operating System (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.03 — Internetwork Operating System (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.03 — Internetwork Operating System (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, M3.03 — Internetwork Operating System (3 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on M3.03 — Internetwork Operating System (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.03 — Internetwork Operating System (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.03 — Internetwork Operating System (3 hours)?
- How would you distinguish M3.03 — Internetwork Operating System (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.03 — Internetwork Operating System (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.03 — Internetwork Operating System (3 hours) incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** M3.03 — Internetwork Operating System (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

### M3.04 — Basic switch configuration (3 hours)

#### Concept and explanation

Basic configuration includes hostname, passwords, management address, default gateway, port settings and saving the configuration. Erasing configuration is a controlled maintenance action.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

#### Study points

- Outline switch setup.
- Explain save and erase operations.
- Relate management address to administration.

## Handbook treatment

M3.04 — Basic switch configuration (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M3.04 — Basic switch configuration (3 hours) using the official terminology retained above.
- Organise the important elements of M3.04 — Basic switch configuration (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.04 — Basic switch configuration (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.04 — Basic switch configuration (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.04 — Basic switch configuration (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M3.04 — Basic switch configuration (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M3.04 — Basic switch configuration (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.04 — Basic switch configuration (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.04 — Basic switch configuration (3 hours)?
- How would you distinguish M3.04 — Basic switch configuration (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.04 — Basic switch configuration (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.04 — Basic switch configuration (3 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M3.04 — Basic switch configuration (3 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

## M3.05 — Basic router configuration (4 hours)

### Concept and explanation

Router configuration sets hostname, interfaces, IP addresses, enable authentication and routes. Copy and TFTP procedures support transfer and backup of configurations.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□□ □□□□□□□□□□.

### Study points

- Outline router setup.
- Explain copy and TFTP use.
- Verify interface status.

# Handbook treatment

M3.05 — Basic router configuration (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M3.05 — Basic router configuration (4 hours) using the official terminology retained above.
- Organise the important elements of M3.05 — Basic router configuration (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.05 — Basic router configuration (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.05 — Basic router configuration (4 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.05 — Basic router configuration (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M3.05 — Basic router configuration (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M3.05 — Basic router configuration (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.05 — Basic router configuration (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.05 — Basic router configuration (4 hours)?
- How would you distinguish M3.05 — Basic router configuration (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.05 — Basic router configuration (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.05 — Basic router configuration (4 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M3.05 — Basic router configuration (4 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□.

## M3.06 — Registers, clock and logging (2 hours)

### Concept and explanation

Configuration-register settings affect boot behaviour. Correct clock settings and IOS message logging improve troubleshooting; buffered and trap logging store or forward events.

**Malayalam support:** □ □□□□□ □□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□ □□□□□□□□.

### Study points

- State register purpose.
- Explain clock and logging.
- Differentiate buffered and trap logging.

## Handbook treatment

M3.06 — Registers, clock and logging (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M3.06 — Registers, clock and logging (2 hours) using the official terminology retained above.
- Organise the important elements of M3.06 — Registers, clock and logging (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M3.06 — Registers, clock and logging (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of IOS, switches and router configuration.
- Write an examination answer on M3.06 — Registers, clock and logging (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M3.06 — Registers, clock and logging (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M3.06 — Registers, clock and logging (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M3.06 — Registers, clock and logging (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M3.06 — Registers, clock and logging (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M3.06 — Registers, clock and logging (2 hours)?
- How would you distinguish M3.06 — Registers, clock and logging (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M3.06 — Registers, clock and logging (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M3.06 — Registers, clock and logging (2 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M3.06 — Registers, clock and logging (2 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

**Module summary:** Configuration is a controlled cycle: inspect, configure, verify, save, back up and monitor.

OFFICIAL MODULE 4

# Static and dynamic routing

Routing selects paths between networks. Static routes are manually configured; dynamic protocols exchange information and calculate paths according to their algorithms and metrics.

**Hours: 14**
**CO4 · Bloom level: Understanding**

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**Handbook study routine:** Complete Module 4 in three passes. First read the official source coverage and topic headings. Next study every Handbook treatment, writing the key definition, relationship, procedure, formula, diagram, or comparison in your own words. Finally close the notes and answer the self-check questions and the module questions without looking at the answer.

## Official source coverage for Module 4

Source basis: Official Contents block. Every point below is retained and linked to a study-oriented explanation. The main topic cards remain the primary lesson narrative.

View extracted official source transcript

Backing up and restoring configurations-upgrading Router's IOS-Disaster Recovery-IOS Authentication and Accounting.
General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route Selection-General Routing Information-Managing Static Routing-Route Control and Redistribution.

## Point-by-point study guide

## Official syllabus point 1: Backing up and restoring configurations-upgrading Router's IOS-Disaster

### Exact source point

**Syllabus text:** Backing up and restoring configurations-upgrading Router's IOS-Disaster

### How to understand and study it

This point is an official syllabus requirement: “Backing up and restoring configurations-upgrading Router's IOS-Disaster”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** ▯ syllabus point ▯▯▯▯▯▯▯▯▯▯▯ Backing up, restoring configurations-upgrading Router's IOS-Disaster ▯▯▯▯ technical terms English-▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯. ▯▯▯▯▯ definition ▯▯▯▯▯▯▯▯▯ principle ▯▯▯▯▯▯▯▯▯▯▯▯; ▯▯▯▯ ▯▯▯ term-▯▯▯▯ function, difference, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Diagram, sequence, formula, unit ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯ revision ▯▯▯▯▯▯▯.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Backing up and restoring configurations-upgrading Router's IOS-Disaster is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Backing up and restoring configurations-upgrading Router's IOS-Disaster using the official terminology retained above.
- Organise the important elements of Backing up and restoring configurations-upgrading Router's IOS-Disaster into a logical sequence, classification, structure, or calculation.
- Connect Backing up and restoring configurations-upgrading Router's IOS-Disaster with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on Backing up and restoring configurations-upgrading Router's IOS-Disaster with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Backing up and restoring configurations-upgrading Router's IOS-Disaster. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Backing up and restoring configurations-upgrading Router's IOS-Disaster is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Backing up and restoring configurations-upgrading Router's IOS-Disaster, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Backing up and restoring configurations-upgrading Router's IOS-Disaster, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Backing up and restoring configurations-upgrading Router's IOS-Disaster?
- How would you distinguish Backing up and restoring configurations-upgrading Router's IOS-Disaster from a closely related idea?
- Where would an engineer or technician encounter Backing up and restoring configurations-upgrading Router's IOS-Disaster, and what must be checked before using it?
- What common mistake would make an answer or operation involving Backing up and restoring configurations-upgrading Router's IOS-Disaster incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Backing up and restoring configurations-upgrading Router's IOS-Disaster [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## Official syllabus point 2: Recovery-IOS Authentication and Accounting.

### Exact source point

**Syllabus text:** Recovery-IOS Authentication and Accounting.

### How to understand and study it

This point is an official syllabus requirement: “Recovery-IOS Authentication and Accounting.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□ Recovery-IOS Authentication, Accounting. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Recovery-IOS Authentication and Accounting. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope Recovery-IOS Authentication and Accounting. using the official terminology retained above.
- Organise the important elements of Recovery-IOS Authentication and Accounting. into a logical sequence, classification, structure, or calculation.
- Connect Recovery-IOS Authentication and Accounting. with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on Recovery-IOS Authentication and Accounting. with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading Recovery-IOS Authentication and Accounting.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of Recovery-IOS Authentication and Accounting. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Recovery-IOS Authentication and Accounting., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Recovery-IOS Authentication and Accounting., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Recovery-IOS Authentication and Accounting.?
- How would you distinguish Recovery-IOS Authentication and Accounting. from a closely related idea?
- Where would an engineer or technician encounter Recovery-IOS Authentication and Accounting., and what must be checked before using it?
- What common mistake would make an answer or operation involving Recovery-IOS Authentication and Accounting. incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** Recovery-IOS Authentication and Accounting. □□□□ topic □□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□.

## Official syllabus point 3: General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing

### Exact source point

> **Syllabus text:** General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing

### How to understand and study it

This point is an official syllabus requirement: “General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing”. Begin with the exact definition or meaning, then identify the purpose, scope, and terminology contained in the point. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□ General Routing Concepts, Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing using the official terminology retained above.
- Organise the important elements of General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing into a logical sequence, classification, structure, or calculation.
- Connect General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing?
- How would you distinguish General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing from a closely related idea?
- Where would an engineer or technician encounter General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing, and what must be checked before using it?
- What common mistake would make an answer or operation involving General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** General Routing Concepts and Terms-TCP/IP Static Routing- TCP/IP Dynamic Routing □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

### Official syllabus point 4: Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route

#### Exact source point

**Syllabus text:** Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route

#### How to understand and study it

This point is an official syllabus requirement: “Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□ Protocols, TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers, Choosing the Right protocol - Route □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route using the official terminology retained above.
- Organise the important elements of Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route into a logical sequence, classification, structure, or calculation.
- Connect Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route. It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route?
- How would you distinguish Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route from a closely related idea?
- Where would an engineer or technician encounter Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route, and what must be checked before using it?
- What common mistake would make an answer or operation involving Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** Protocols: TCP/IP Interior Gateway Protocols-TCP/IP Exterior Gateway Protocols-Configuring IP Routing Protocols on Routers: Choosing the Right protocol - Route □□□□ topic □□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□□ □□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□□ concept-□□□□□ □□□□□□–□□□ □□□□□□ □□□□□□□□□□□□□□□.

## Official syllabus point 5: Selection-General Routing Information-Managing Static Routing-Route Control and

### Exact source point

> **Syllabus text:** Selection-General Routing Information-Managing Static Routing-Route Control and

### How to understand and study it

This point is an official syllabus requirement: “Selection-General Routing Information-Managing Static Routing-Route Control and”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

> **Malayalam support:** □ syllabus point □□□□□□□□□□□□ Selection-General Routing Information-Managing Static Routing-Route Control and □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

> **Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

# Handbook treatment

Selection-General Routing Information-Managing Static Routing-Route Control and should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

## Learning objectives

- Define and scope Selection-General Routing Information-Managing Static Routing-Route Control and using the official terminology retained above.
- Organise the important elements of Selection-General Routing Information-Managing Static Routing-Route Control and into a logical sequence, classification, structure, or calculation.
- Connect Selection-General Routing Information-Managing Static Routing-Route Control and with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on Selection-General Routing Information-Managing Static Routing-Route Control and with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading Selection-General Routing Information-Managing Static Routing-Route Control and. It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

## Engineering application lens

Engineering use of Selection-General Routing Information-Managing Static Routing-Route Control and depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

## Worked answer method

**Given:** the quantity to be sensed, the operating environment, and the required decision or control action.
**Required:** a suitable measurement chain or an explanation of how the instrument operates.
**Method:** map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
**Answer:** state the measured quantity, principle, important characteristics, application, and limitation.

### Exam answer blueprint

For a short-answer question on Selection-General Routing Information-Managing Static Routing-Route Control and, begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Selection-General Routing Information-Managing Static Routing-Route Control and, and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Selection-General Routing Information-Managing Static Routing-Route Control and?
- How would you distinguish Selection-General Routing Information-Managing Static Routing-Route Control and from a closely related idea?
- Where would an engineer or technician encounter Selection-General Routing Information-Managing Static Routing-Route Control and, and what must be checked before using it?
- What common mistake would make an answer or operation involving Selection-General Routing Information-Managing Static Routing-Route Control and incorrect?

### Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

**Malayalam support:** Selection-General Routing Information-Managing Static Routing-Route Control and □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

### Official syllabus point 6: Redistribution.

#### Exact source point

**Syllabus text:** Redistribution.

#### How to understand and study it

This point is an official syllabus requirement: “Redistribution.”. Treat every named term as an examinable subpoint. Define it, explain its role or behaviour, distinguish it from related terms, and connect it to the module application. The main topic explanation above gives the concept context; this point-by-point section ensures that each named item is revised separately.

#### Study sequence

1. Write the exact technical term and a one-sentence definition in your own words.
2. List the named terms separately and add one distinguishing feature or engineering use for each.
3. Finish with one examination answer: definition, explanation, diagram/table if relevant, and application or limitation.

**Malayalam support:** □ syllabus point □□□□□□□□□□□ Redistribution. □□□□ technical terms English-□□□□□□□ □□□□□□□. □□□□□ definition □□□□□□□□□□ principle □□□□□□□□□□□□□; □□□□ □□□ term-□□□□ function, difference, application □□□□□□ □□□□□□□□□□ □□□□□□. Diagram, sequence, formula, unit □□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□ revision □□□□□□□.

**Exam focus:** Answer this point with its definition or principle first, followed by the named features, sequence, comparison, formula, diagram, or application that the question demands.

## Handbook treatment

Redistribution. is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope Redistribution. using the official terminology retained above.
- Organise the important elements of Redistribution. into a logical sequence, classification, structure, or calculation.
- Connect Redistribution. with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on Redistribution. with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading Redistribution.. It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of Redistribution. is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on Redistribution., begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is Redistribution., and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining Redistribution.?
- How would you distinguish Redistribution. from a closely related idea?
- Where would an engineer or technician encounter Redistribution., and what must be checked before using it?
- What common mistake would make an answer or operation involving Redistribution. incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

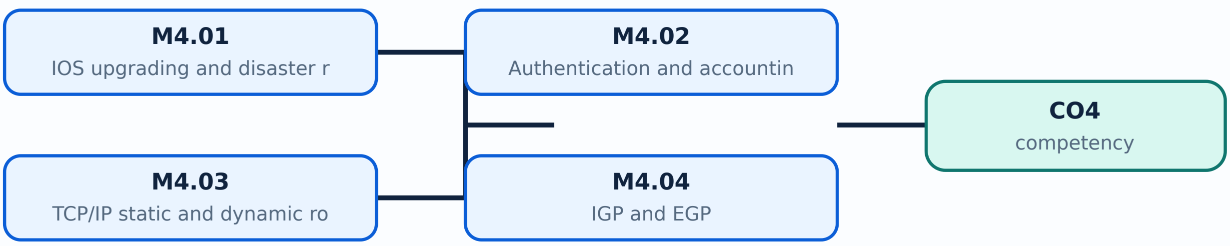
**Malayalam support:** Redistribution. □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

## Learning goals

- Explain IOS recovery and authentication.
- Configure concepts of static and dynamic routing.
- Compare IGP, EGP and route control.

## Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.




Learning map for Module module-4: the listed topics build the module competency.

## M4.01 — IOS upgrading and disaster recovery (2 hours)

### Concept and explanation

IOS images and configuration files must be backed up before upgrade. Disaster recovery includes boot behaviour, image selection, configuration restoration and documented rollback.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

### Study points

- List upgrade precautions.
- Explain recovery sequence.
- Identify configuration backup needs.

## Handbook treatment

M4.01 — IOS upgrading and disaster recovery (2 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M4.01 — IOS upgrading and disaster recovery (2 hours) using the official terminology retained above.
- Organise the important elements of M4.01 — IOS upgrading and disaster recovery (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.01 — IOS upgrading and disaster recovery (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.01 — IOS upgrading and disaster recovery (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M4.01 — IOS upgrading and disaster recovery (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M4.01 — IOS upgrading and disaster recovery (2 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M4.01 — IOS upgrading and disaster recovery (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.01 — IOS upgrading and disaster recovery (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.01 — IOS upgrading and disaster recovery (2 hours)?
- How would you distinguish M4.01 — IOS upgrading and disaster recovery (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.01 — IOS upgrading and disaster recovery (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.01 — IOS upgrading and disaster recovery (2 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M4.01 — IOS upgrading and disaster recovery (2 hours)
□□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

## M4.02 — Authentication and accounting (1 hours)

### Concept and explanation

Authentication verifies identity; authorisation controls permitted actions; accounting records activity. These controls support accountability in network administration.

> **Malayalam support:** □ □□□□□ □□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□ □□□□□□□□.

### Study points

- Define AAA concepts.
- Relate authentication to device access.

# Handbook treatment

M4.02 — Authentication and accounting (1 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

## Learning objectives

- Define and scope M4.02 — Authentication and accounting (1 hours) using the official terminology retained above.
- Organise the important elements of M4.02 — Authentication and accounting (1 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.02 — Authentication and accounting (1 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.02 — Authentication and accounting (1 hours) with a clear opening, technical middle, and final application or limitation.

## Conceptual framework

Use the following frame while reading M4.02 — Authentication and accounting (1 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

## Engineering application lens

The engineering value of M4.02 — Authentication and accounting (1 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

## Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M4.02 — Authentication and accounting (1 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.02 — Authentication and accounting (1 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.02 — Authentication and accounting (1 hours)?
- How would you distinguish M4.02 — Authentication and accounting (1 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.02 — Authentication and accounting (1 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.02 — Authentication and accounting (1 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M4.02 — Authentication and accounting (1 hours) □□□□□ topic □□□□□□□□□□□□□□□ □□□□□□ exact technical term □□□□□□□□□□□□□□□□. □□□□□ definition □□□□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□□ □□□□□□□□□□ □□□□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□□□ □□□□□□□□□□□. Exam answer □□□□□□□□□□□□□ concept-□□□□□ □□□□□□□–□□□□ □□□□□□ □□□□□□□□□□□□□□□□.

## M4.03 — TCP/IP static and dynamic routing (3 hours)

### Concept and explanation

A static route is manually specified and predictable but requires maintenance. Dynamic routing learns paths through protocols and adapts to topology changes.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□□ □□□□□□□□□.

### Study points

- Compare static and dynamic routing.
- Explain next hop and route selection.
- Identify suitable use cases.

## Handbook treatment

M4.03 — TCP/IP static and dynamic routing (3 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M4.03 — TCP/IP static and dynamic routing (3 hours) using the official terminology retained above.
- Organise the important elements of M4.03 — TCP/IP static and dynamic routing (3 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.03 — TCP/IP static and dynamic routing (3 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.03 — TCP/IP static and dynamic routing (3 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M4.03 — TCP/IP static and dynamic routing (3 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M4.03 — TCP/IP static and dynamic routing (3 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M4.03 — TCP/IP static and dynamic routing (3 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.03 — TCP/IP static and dynamic routing (3 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.03 — TCP/IP static and dynamic routing (3 hours)?
- How would you distinguish M4.03 — TCP/IP static and dynamic routing (3 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.03 — TCP/IP static and dynamic routing (3 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.03 — TCP/IP static and dynamic routing (3 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M4.03 — TCP/IP static and dynamic routing (3 hours) [illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M4.04 — IGP and EGP (4 hours)

### Concept and explanation

Interior Gateway Protocols operate within an autonomous system; Exterior Gateway Protocols exchange routing information between autonomous systems. Protocol choice depends on scale, policy and convergence requirements.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□□ □□□□□□□□□□.

### Study points

- Differentiate IGP and EGP.
- State routing-domain concepts.
- Relate policy to protocol choice.

## Handbook treatment

M4.04 — IGP and EGP (4 hours) is an official examinable part of the module. Study it as a connected explanation: define the central term, identify the related elements, explain their relationship, and finish with a relevant engineering use or limitation.

### Learning objectives

- Define and scope M4.04 — IGP and EGP (4 hours) using the official terminology retained above.
- Organise the important elements of M4.04 — IGP and EGP (4 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.04 — IGP and EGP (4 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.04 — IGP and EGP (4 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M4.04 — IGP and EGP (4 hours). It is a study organisation method, not an additional official syllabus requirement.

- Start with the exact technical term and its scope.
- Separate the topic into named elements, stages, types, or conditions.
- Explain the relationship using cause-and-effect, sequence, comparison, or a labelled representation.
- Apply the idea to a realistic engineering situation and state one limitation or common mistake.

### Engineering application lens

The engineering value of M4.04 — IGP and EGP (4 hours) is understood by asking what requirement it satisfies, what input or condition it depends on, how the output is obtained, and how a technician or designer verifies the result. Use the module context to make the application specific rather than memorising an isolated definition.

### Worked answer method

**Given:** the topic, operating condition, diagram, data, or situation stated in the question.
**Required:** definition, explanation, comparison, procedure, calculation, or application as requested.
**Method:** organise the answer under principle, named elements, sequence or relationship, and application.
**Answer:** use the requested technical terms, units, labels, assumptions, and a concluding statement.

### Exam answer blueprint

For a short-answer question on M4.04 — IGP and EGP (4 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.04 — IGP and EGP (4 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.04 — IGP and EGP (4 hours)?
- How would you distinguish M4.04 — IGP and EGP (4 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.04 — IGP and EGP (4 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.04 — IGP and EGP (4 hours) incorrect?

### Common mistakes and safety emphasis

- Copying a definition without explaining the mechanism or purpose.
- Listing terms without showing their relationship.
- Using an example that does not match the stated topic or condition.
- Leaving out a diagram, unit, assumption, limitation, or safety point when the question requires it.

**Malayalam support:** M4.04 — IGP and EGP (4 hours) □□□□ topic □□□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□□.

## M4.05 — Protocol choice and routing information (2 hours)

### Concept and explanation

Route selection uses destination, prefix length, administrative preference, metric and protocol information. A suitable protocol balances convergence, resource use and operational complexity.

> **Malayalam support:** [illegible] English technical terms, symbols, units, equations, and standard abbreviations [illegible].

### Study points

- Explain route-selection factors.
- Interpret routing information conceptually.

## Handbook treatment

M4.05 — Protocol choice and routing information (2 hours) should be understood as an information-flow problem. Identify the input, the rule or program that processes it, the internal state or decision, and the output. A correct explanation must show the sequence, not merely list software terms.

### Learning objectives

- Define and scope M4.05 — Protocol choice and routing information (2 hours) using the official terminology retained above.
- Organise the important elements of M4.05 — Protocol choice and routing information (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.05 — Protocol choice and routing information (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.05 — Protocol choice and routing information (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M4.05 — Protocol choice and routing information (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- List the input signals, data, or conditions.
- Write the logic, algorithm, instruction sequence, or protocol rule in the order it is evaluated.
- State the output and identify what changes when an input or state changes.
- Test a normal case, a boundary case, and a fault or invalid-input case.

### Engineering application lens

For engineering practice, M4.05 — Protocol choice and routing information (2 hours) matters because an automated or digital system must convert inputs into repeatable outputs. The technician should be able to trace a signal or data item, locate the decision that controls the result, and identify a safe response when an input is missing or abnormal.

### Worked answer method

**Given:** the input conditions, required output, and any stated constraints.
**Required:** the logic sequence, program structure, truth table, or operating result.
**Method:** break the problem into named states or steps; evaluate them in order; test at least one alternate condition.
**Answer:** show the final sequence and state the expected output for each important condition.

### Exam answer blueprint

For a short-answer question on M4.05 — Protocol choice and routing information (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.05 — Protocol choice and routing information (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.05 — Protocol choice and routing information (2 hours)?
- How would you distinguish M4.05 — Protocol choice and routing information (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.05 — Protocol choice and routing information (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.05 — Protocol choice and routing information (2 hours) incorrect?

### Common mistakes and safety emphasis

- Writing instructions without defining the input and output.
- Ignoring scan order, state retention, initialization, or boundary conditions.
- Confusing a logical condition with the physical signal that represents it.
- Testing only the normal case and failing to consider a fault or interlock condition.

**Malayalam support:** M4.05 — Protocol choice and routing information (2 hours)
[illegible] topic [illegible] [illegible] exact technical term [illegible]. [illegible] definition [illegible] principle, named parts/steps, application, limitation [illegible] [illegible] [illegible]. English terminology, symbols, units, diagram labels [illegible] [illegible]. Exam answer [illegible] concept-[illegible] [illegible]–[illegible] [illegible] [illegible].

## M4.06 — Route control and redistribution (2 hours)

### Concept and explanation

Route control filters or modifies routing information. Redistribution exchanges routes between protocols but must be designed to avoid loops, inconsistent metrics and unintended reachability.

> **Malayalam support:** □ □□□□□□ □□□□□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□□□ □□□□□□□□□.

### Study points

- Define redistribution.
- State route-control purposes.
- Identify loop and policy risks.

## Handbook treatment

M4.06 — Route control and redistribution (2 hours) should be learned through the chain of measurand, sensing element, signal conversion, indication or processing, and decision. The purpose of every stage is more important than memorising a component name alone.

### Learning objectives

- Define and scope M4.06 — Route control and redistribution (2 hours) using the official terminology retained above.
- Organise the important elements of M4.06 — Route control and redistribution (2 hours) into a logical sequence, classification, structure, or calculation.
- Connect M4.06 — Route control and redistribution (2 hours) with one engineering decision, operation, measurement, or safety requirement in the context of Static and dynamic routing.
- Write an examination answer on M4.06 — Route control and redistribution (2 hours) with a clear opening, technical middle, and final application or limitation.

### Conceptual framework

Use the following frame while reading M4.06 — Route control and redistribution (2 hours). It is a study organisation method, not an additional official syllabus requirement.

- Define the physical quantity or measurand.
- Identify the sensing or conversion element and the form of output it produces.
- Explain conditioning, transmission, indication, or control if present.
- Discuss range, sensitivity, accuracy, repeatability, response, calibration, and limitations where relevant.

### Engineering application lens

Engineering use of M4.06 — Route control and redistribution (2 hours) depends on whether the measurement is sufficiently accurate, fast, repeatable, robust, and compatible with the controller or display. The selection must also consider installation, environment, maintenance, and failure mode.

### Worked answer method

**Given:** the quantity to be sensed, the operating environment, and the required decision or control action.
**Required:** a suitable measurement chain or an explanation of how the instrument operates.
**Method:** map measurand → sensing element → signal → processing/indication → action; identify one source of error and one calibration check.
**Answer:** state the measured quantity, principle, important characteristics, application, and limitation.

### Exam answer blueprint

For a short-answer question on M4.06 — Route control and redistribution (2 hours), begin with the definition or principle and include the decisive feature. For a medium-answer question, add the named elements and their functions or sequence. For a long-answer question, add a labelled diagram, comparison, formula or procedure where relevant, one application, and one limitation or safety point. Never substitute a list of headings for the explanation.

### Self-check questions

- What is M4.06 — Route control and redistribution (2 hours), and what is its scope in this module?
- Which elements, stages, types, quantities, or conditions must be named when explaining M4.06 — Route control and redistribution (2 hours)?
- How would you distinguish M4.06 — Route control and redistribution (2 hours) from a closely related idea?
- Where would an engineer or technician encounter M4.06 — Route control and redistribution (2 hours), and what must be checked before using it?
- What common mistake would make an answer or operation involving M4.06 — Route control and redistribution (2 hours) incorrect?

### Common mistakes and safety emphasis

- Confusing accuracy, precision, sensitivity, and resolution.
- Calling every electrical output a direct measurement without explaining conversion.
- Ignoring calibration, loading, response time, or environmental effects.
- Failing to state the unit and reference condition of the measurand.

**Malayalam support:** M4.06 — Route control and redistribution (2 hours) □□□□ topic □□□□□□□□□□□□ □□□□□ exact technical term □□□□□□□□□□□□□. □□□□ definition □□□□□□□□□□ principle, named parts/steps, application, limitation □□□□□□ □□□□□□□□ □□□□□□□□. English terminology, symbols, units, diagram labels □□□□□□ □□□□□□□□□. Exam answer □□□□□□□□□□ concept-□□□□ □□□□□–□□□ □□□□□ □□□□□□□□□□□□□.

**Module summary:** Routing knowledge connects address planning, device configuration, protocol operation and network policy.

## Formula and Notation Bank

**Formula note:** The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

## Diagram Library for Rapid Revision

Open the module to view its labelled learning-map diagram.

**Module 2: Active Directory**

Open the module to view its labelled learning-map diagram.

**Module 3: IOS, switches and**

Open the module to view its labelled learning-map diagram.

**Module 4: Static and**

Open the module to view its labelled learning-map diagram.

## Official Activities and Practical Requirements

### Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

### Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

## Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied syllabus lists Series Test-1 and Series Test-2 as 1-hour entries and does not state a separate CIE/ESE mark distribution.

## Module-wise Questions and Answers

### module-1 — Network control devices, wireless and services

**1. Explain IPv4 and IPv6 addressing.** **[3 marks]**

IPv4 uses 32-bit addresses and subnet masks; IPv6 uses 128-bit addresses and a larger address space. Prefix notation identifies the network portion in IPv6 and modern IPv4 practice.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**2. Explain Network-control devices.** **[3 marks]**

Hubs repeat signals, switches forward frames using MAC addresses, routers forward packets between networks, repeaters regenerate signals, VPN devices create protected tunnels and modems adapt signals to a service.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**3. Explain Wireless technologies.** **[3 marks]**

Wi-Fi, Bluetooth and WiMAX provide wireless connectivity with different range, data rate, power and topology characteristics. Security and interference must be considered.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**4. Explain IEEE 802 standards. [3 marks]**

IEEE 802 standards define families of LAN and wireless networking technologies. The exact standard determines medium access, physical layer and interoperability requirements.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## module-2 — Active Directory and security

**5. Explain Domain controllers and Active Directory. [3 marks]**

A domain controller authenticates users and applies directory-based administration. Active Directory stores objects, accounts, groups and policy information in a structured domain.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**6. Explain Domain and workgroup users. [3 marks]**

A workgroup manages accounts locally, while a domain centralises authentication and policy. User and group accounts, memberships, child domains and additional domain controllers support administration and redundancy.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**7. Explain File and folder security. [3 marks]**

Share permissions and security permissions control access at different layers. Effective access depends on the combined rules, inheritance, ownership and principle of least privilege.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**8. Explain Security, firewall and backup.** **[3 marks]**

A firewall filters traffic according to rules and network context. Encryption protects data confidentiality; system and Active Directory backups support recovery from failure or attack.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## module-3 — IOS, switches and router configuration

**9. Explain Switch and router hardware and memory.** **[3 marks]**

Network devices contain processors, interfaces and memory areas such as running configuration, startup configuration and IOS storage. Understanding memory explains why changes may disappear after reboot.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**10. Explain Router and switch interfaces.** **[3 marks]**

Interfaces connect a device to network media and require correct speed, duplex, addressing, status and descriptions. Switch ports and router interfaces have different forwarding roles.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**11. Explain Internetwork Operating System.** **[3 marks]**

IOS provides command modes, configuration commands, show commands and copy operations. Command hierarchy controls access to monitoring and configuration functions.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**12. Explain Basic switch configuration.** **[3 marks]**

Basic configuration includes hostname, passwords, management address, default gateway, port settings and saving the configuration. Erasing configuration is a controlled maintenance action.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## module-4 — Static and dynamic routing

**13. Explain IOS upgrading and disaster recovery.** **[3 marks]**

IOS images and configuration files must be backed up before upgrade. Disaster recovery includes boot behaviour, image selection, configuration restoration and documented rollback.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**14. Explain Authentication and accounting.** **[3 marks]**

Authentication verifies identity; authorisation controls permitted actions; accounting records activity. These controls support accountability in network administration.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**15. Explain TCP/IP static and dynamic routing.** **[3 marks]**

A static route is manually specified and predictable but requires maintenance. Dynamic routing learns paths through protocols and adapts to topology changes.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

**16. Explain IGP and EGP.** **[3 marks]**

Interior Gateway Protocols operate within an autonomous system; Exterior Gateway Protocols exchange routing information between autonomous systems. Protocol choice depends on scale, policy and convergence requirements.

**Exam focus:** State the definition or purpose, explain the working or sequence, and add one application or caution.

## Practice Model Question Paper

### Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *IPv4 and IPv6 addressing* with one relevant application. (5 marks)
2. **2.** Define or explain *Network-control devices* with one relevant application. (5 marks)
3. **3.** Define or explain *Domain controllers and Active Directory* with one relevant application. (5 marks)
4. **4.** Define or explain *Domain and workgroup users* with one relevant application. (5 marks)
5. **5.** Define or explain *Switch and router hardware and memory* with one relevant application. (5 marks)
6. **6.** Define or explain *Router and switch interfaces* with one relevant application. (5 marks)
7. **7.** Define or explain *IOS upgrading and disaster recovery* with one relevant application. (5 marks)
8. **8.** Define or explain *Authentication and accounting* with one relevant application. (5 marks)

> **Writing advice:** Use headings, standard terminology, and labelled diagrams or procedures where relevant.

## Quick Revision

### Module-wise key points

- **Network control devices, wireless and services:** Network operation depends on correctly combining addresses, devices, standards and automatic configuration services.

- **Active Directory and security:** Identity, permissions and recovery controls protect network resources and users.
- **IOS, switches and router configuration:** Configuration is a controlled cycle: inspect, configure, verify, save, back up and monitor.
- **Static and dynamic routing:** Routing knowledge connects address planning, device configuration, protocol operation and network policy.

### Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

**Last-minute revision:** Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

# Glossary

#### Technical glossary

| Term | Short meaning |
|---|---|
| **IPv4 and IPv6 addressing** | IPv4 uses 32-bit addresses and subnet masks; IPv6 uses 128-bit addresses and a larger address space. Prefix notation identifies the network portion in IPv6 and modern IPv4 practice. |
| **Network-control devices** | Hubs repeat signals, switches forward frames using MAC addresses, routers forward packets between networks, repeaters regenerate signals, VPN devices create protected tunnels and modems adapt signals to a service. |
| **Wireless technologies** | Wi-Fi, Bluetooth and WiMAX provide wireless connectivity with different range, data rate, power and topology characteristics. Security and interference must be considered. |
| **IEEE 802 standards** | IEEE 802 standards define families of LAN and wireless networking technologies. The exact standard determines medium access, physical layer and interoperability requirements. |
| **DNS and DHCP** | DNS resolves names through a distributed hierarchy. DHCP leases configuration such as IP address, mask, gateway and DNS server. Installation requires correct server roles, scopes, records and client settings. |

| Term | Short meaning |
|---|---|
| **Domain controllers and Active Directory** | A domain controller authenticates users and applies directory-based administration. Active Directory stores objects, accounts, groups and policy information in a structured domain. |
| **Domain and workgroup users** | A workgroup manages accounts locally, while a domain centralises authentication and policy. User and group accounts, memberships, child domains and additional domain controllers support administration and redundancy. |
| **File and folder security** | Share permissions and security permissions control access at different layers. Effective access depends on the combined rules, inheritance, ownership and principle of least privilege. |
| **Security, firewall and backup** | A firewall filters traffic according to rules and network context. Encryption protects data confidentiality; system and Active Directory backups support recovery from failure or attack. |
| **Switch and router hardware and memory** | Network devices contain processors, interfaces and memory areas such as running configuration, startup configuration and IOS storage. Understanding memory explains why changes may disappear after reboot. |
| **Router and switch interfaces** | Interfaces connect a device to network media and require correct speed, duplex, addressing, status and descriptions. Switch ports and router interfaces have different forwarding roles. |
| **Internetwork Operating System** | IOS provides command modes, configuration commands, show commands and copy operations. Command hierarchy controls access to monitoring and configuration functions. |
| **Basic switch configuration** | Basic configuration includes hostname, passwords, management address, default gateway, port settings and saving the configuration. Erasing configuration is a controlled maintenance action. |
| **Basic router configuration** | Router configuration sets hostname, interfaces, IP addresses, enable authentication and routes. Copy and TFTP procedures support transfer and backup of configurations. |
| **Registers, clock and logging** | Configuration-register settings affect boot behaviour. Correct clock settings and IOS message logging improve troubleshooting; buffered and trap logging store or forward events. |
| **IOS upgrading and disaster recovery** | IOS images and configuration files must be backed up before upgrade. Disaster recovery includes boot behaviour, image selection, configuration restoration and documented rollback. |
| **Authentication and accounting** | Authentication verifies identity; authorisation controls permitted actions; accounting records activity. These controls support accountability in network administration. |
| **TCP/IP static and dynamic routing** | A static route is manually specified and predictable but requires maintenance. Dynamic routing learns paths through protocols and adapts to topology changes. |
| **IGP and EGP** | Interior Gateway Protocols operate within an autonomous system; Exterior Gateway Protocols exchange routing information between autonomous systems. Protocol choice depends on scale, policy and convergence requirements. |

| Term | Short meaning |
|---|---|
| **Protocol choice and routing information** | Route selection uses destination, prefix length, administrative preference, metric and protocol information. A suitable protocol balances convergence, resource use and operational complexity. |
| **Route control and redistribution** | Route control filters or modifies routing information. Redistribution exchanges routes between protocols but must be designed to avoid loops, inconsistent metrics and unintended reachability. |

# Official References and Resources

## References listed in the supplied syllabus


1. Computer Networks, Andrew S. Tanenbaum.
2. Understanding the Network: A Practical Guide to Internetworking, Michael J. Martin.


## Official learning resources listed

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# In-page Search

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